

Hernani Samuel Diniz

Java Backend Developer — Spring Boot · REST APIs · PostgreSQL

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Itaúna, MG, Brazil (open to remote worldwide · open to relocation for international roles)

SUMMARY

Recent Software Engineering graduate specializing in Java and Spring Boot, seeking a first backend role. Built a production-grade REST API (Munehisa) with layered architecture, JWT authentication, and a three-stage automated test suite in CI, demonstrating professional-level engineering practice through independent projects. Background in low-level systems programming (Rust) adds a strong foundation in state modeling and correctness discipline to backend design.

TECHNICAL SKILLS

Languages: Java, Python, Rust, C, TypeScript, SQL

Backend & APIs: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Bean Validation, RESTful API design, JWT authentication, OpenAPI/Swagger

Data: PostgreSQL, Flyway schema migrations

Testing: JUnit 5, Mockito, Testcontainers, k6 load testing

DevOps & Tools: Docker, Docker Compose, GitHub Actions (CI/CD), Maven, Git

Architecture: Layered (controller/service/repository) design, microservices, Architecture Decision Records

TECHNICAL PROJECTS

Munehisa Investment Simulator [GitHub](#) | [Live demo](#) | Jun 2026 – Aug 2026

Java 21 · Spring Boot 4.1 · PostgreSQL 16 · React 19/TypeScript · Python/FastAPI

- Designed and built a **25-endpoint REST API** in Java 21 / Spring Boot 4.1 with a layered controller → service → repository architecture.
- Implemented **JWT-based stateless authentication** (Spring Security) with email verification, login, and password reset; modeled trading, deposits/withdrawals, and inflation-adjusted transactions in the service layer.
- PostgreSQL 16** with Flyway migrations; **51 test classes** (JUnit 5/Mockito + Testcontainers) in a **3-stage GitHub Actions CI pipeline**, plus a k6 load-testing suite (5 scenarios).
- Split market-data fetching into a separate **Python/FastAPI microservice**; documented **12 ADRs** and **15 Business Rule Records**; deployed on a Hetzner Cloud VM with Swagger UI docs.

RISC-V RV32IMA Emulator [GitHub](#) | [Live demo](#) — Linux in the browser | Mar 2026 – May 2026

Rust · WebAssembly

- Built a full RV32IMA emulator in Rust (integer, multiply/divide, atomic instructions, machine-mode privileged spec) that boots a real **Linux 6.1.14 kernel** and FreeRTOS bare-metal images.
- Modeled CPU/trap state with Rust's type system to make invalid states unrepresentable; built own disassembler/tracer/debugger tooling; compiles to WebAssembly to run client-side in the browser.

EDUCATION & LANGUAGES

B.Sc. in Software Engineering — Unicesumar (Feb 2021 – Feb 2025)

Portuguese (native) · English — advanced technical reading (ISA specs, RFCs, official documentation)